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ORIGINAL ARTICLE

Three-dimensional gravity inversion based on the Hamiltonian Monte Carlo method

Ya Xu[1,2] | **Wei Chu**[2,1] | **Song Huang**[1] | **Rui Guo**[3] | **Shupeng Lu**[1,2] | **Fangzhou Nan**[1]

[1]Key Laboratory of Deep Petroleum Intelligent Exploration and Development, Institute of Geology and Geophysics, Chinese Academy of Sciences, Beijing, China

[2]College of Earth and Planetary Sciences, University of Chinese Academy of Sciences, Beijing, China

[3]Exploration Surveying Institute of Baogang Group, Baotou, China

Correspondence
Ya Xu, Key Laboratory of Deep Petroleum Intelligent Exploration and Development, Institute of Geology and Geophysics, Chinese Academy of Sciences, Beijing, China. Email: xuya@mail.iggcas.ac.cn

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Abstract

In geophysics, Bayesian inversion methods are of significant prominence. Here, we present a novel approach utilizing the Hamiltonian Monte Carlo (HMC) method in gravity inversion for elucidating three-dimensional (3D) density structures. HMC provides a multi-dimensional sampling method that demonstrates enhanced optimization efficiency, facilitating the attainment of distant proposals with elevated acceptance probabilities. Its applicability also extends to resolving linear inverse problems. Three synthetic models of cubic bodies, dipping dykes and a combined model were designed for tests. The testes underscore the promising potential of HMC in recovering subsurface density source bodies and giving the uncertainty of the inversion model. Furthermore, an inversion test conducted on the Vinton salt dome yields a reasonable 3D distribution of cap rock, consistent with prior studies in this area. The modelling and field experiments showed that the proposed HMC gravity inversion method had higher accuracy and application potential.

KEYWORDS
gravity, inversion, potential field

INTRODUCTION

Gravity inversion is a conventional geophysical technique used for estimating subsurface mass density distribution (Li & Oldenburg, 1998). It is extensively employed in mineral and hydrocarbon exploration and tectonic studies (Hinze et al., 2013). The gravity inversion problem can be expressed as a linear relationship between gravity observations and densities of a discretized array of subsurface rectangular prisms. However, due to limited observations and noisy data, the gravity inverse problem inherently presents non-uniqueness (Backus & Gilbert, 1970; Zhdanov, 2015). Regularization methods are commonly introduced to address such an ill-posed inverse problems (Tikhonov & Arsenin, 1977), yielding specific optimal solutions within a framework of acceptable fitting errors and model constraints. Nonetheless, within this framework, global optimality may not be ensured and uncertainty assessment of the inversion result remains challenging due to limited information (Peng et al., 2022).

Another prevalent approach that deals with the inversion problem involves probabilistic methods, which furnish information through probability density functions across the model parameter space. Bayesian algorithms, portraying the inversion problem in terms of posterior probability density function, stand as a notable instance. For instance, the Monte Carlo (MC) method, facilitating direct simulation of probability distributions, has emerged as a prominent geophysical inversion method in the past decades (Mosegaard & Sambridge, 2002). It is widely used in large-scale, multi-parameter inversion problems in geophysical studies (Afonso, Fullea, Griffin et al., 2013; Afonso, Fullea, Yang et al., 2013; Afonso et al., 2016). Bosch et al. (2006) notably applied the MC method to joint gravity and magnetic inversion. The advancement of Markov Chain Monte Carlo (MCMC) engenders a specialized

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sequential process for random samples, laying the groundwork for generating subsequent random samples. Particularly valuable for obtaining posterior distributions in analytically challenging scenarios, MCMC mitigates reliance on the initial model and furnishes uncertainty analysis for inversion results through model space exploration (van Ravenzwaaij et al., 2018; Wei et al., 2023). Multi-dimensional random sample generation poses a key challenge for all MC techniques, with the Metropolis-Hasting (MH) algorithm serving as the foundational algorithm for MCMC inversion, upon which numerous sampling algorithms are built.

In recent years, the Hamiltonian Monte Carlo (HMC) method has emerged as a novel sampling mechanism (Betancourt et al., 2017; Duane et al., 1987; Neal, 2011). HMC generates proposal distributions by introducing an auxiliary variable corresponding to the particles' momentum in accordance with Hamiltonian principles. This method adeptly navigates state space, reaching distant proposals with high acceptance probabilities within an MHs framework, thereby facilitating more efficient exploration compared to traditional MCMC samplers such as random walk MHs(Neal 2011). Sen and Biswas (2017) pioneered its application in geophysical contexts, particularly in acoustic impedance inversion. HMC finds extensive utility in seismic source inversion (Fichtner & Simute, 2018), tomography (Fichtner & Zunino, 2019; Fichtner et al., 2019; Muir & Tkalčić, 2020), elastic modulus estimation (Koch et al., 2019), amplitude versus angle inversion (Aleardi & Salusti, 2020), Rayleigh-wave dispersion inversion (Aleardi et al., 2020), two-dimensional (2D) full-waveform inversion (Gebraad et al., 2020), and co-seismic fault model estimation (Yamada et al., 2022). Additionally, HMC methods have been effectively applied to gravity and magnetic inversion for profile data (Zunino et al., 2022, 2023), further underlining its promise in addressing geophysical inversion challenges.

This study proposes a 3D gravity inversion method founded on the HMC algorithm. Initially, we delineate the fundamental theory underlying the HMC gravity inversion method. Subsequently, we validate our algorithm through synthetic model testing. Finally, we apply this method to a field example, showcasing its practical efficacy.

METHODS

Inversion problem

The subsurface structures can be represented by discrete prismatic units. For the density structure, we discretize the subsurface into a set of prisms with density contrasts as the model parameter **m**. The gravity forward modelling constitutes a linear problem, and it can be expressed in matrix notation $\mathbf{d} = \mathbf{Am}$, wherein the relationship between the gravity anomaly data **d** and the model parameter **m** with a forward kernel matrix **A** (Blakely, 1995).

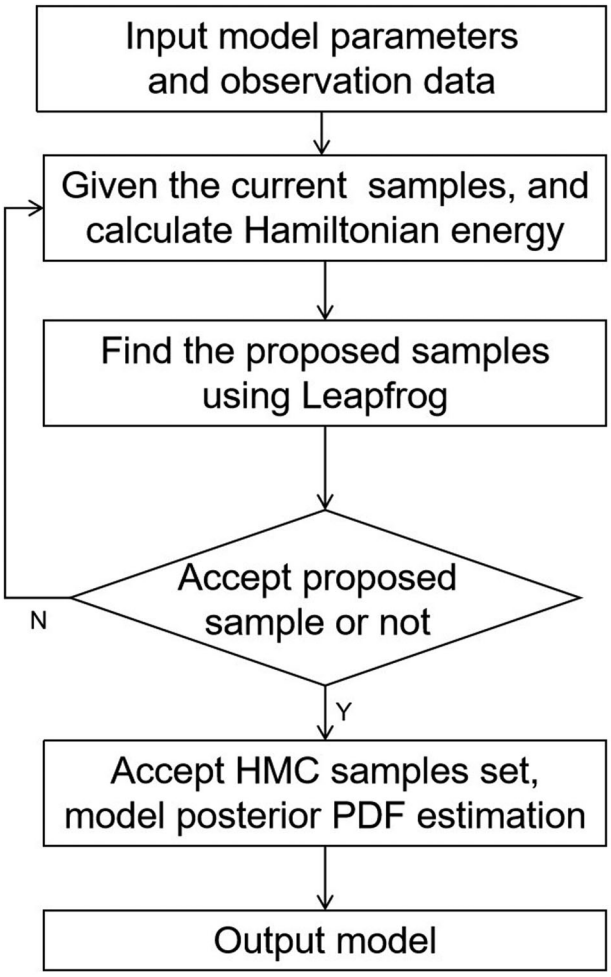


FIGURE 1 Flow diagram of Hamiltonian Monte Carlo (HMC) gravity inversion algorithm.

In a probabilistic inversion framework, the inversion is equivalent to solving the posterior probability density function using Bayes' theorem, also recognized as the Bayesian inference problem (Tarantola, 2005). According to the Bayes rule, the PDF (probability density function) function can be expressed as

$$\mathbf{P}(\mathbf{m}|\mathbf{d}) = \frac{\mathbf{P}(\mathbf{d}|\mathbf{m})\,\mathbf{P}(\mathbf{m})}{\mathbf{P}(\mathbf{d})} \propto \mathbf{P}(\mathbf{d}|\mathbf{m})\,\mathbf{P}(\mathbf{m}), \qquad (1)$$

where **P**(**m**) is the prior distribution representing the information on the density model, **P**(**d**|**m**) is the likelihood function, **P**(**d**) is the evidence and generally taken as a constant. **P**(**m**|**d**) is the posterior PDF, and it is proportional to the likelihood PDF times the prior PDF. The model **m** that maximizes the PDF (Equation 1) is the Maximum A Posteriori estimate.

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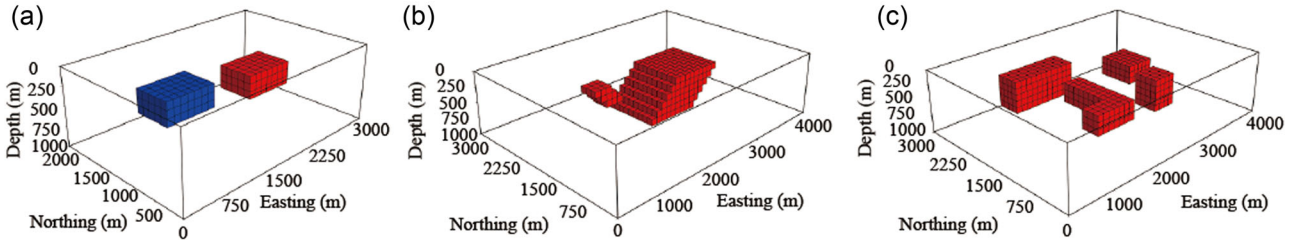


FIGURE 2 The synthetic models. (a) Model 1 with two cubic bodies (b) Model 2 with two dipping dykes and (c) Model 3 with four source bodies.

TABLE 1 Geometry parameters of anomaly bodies in the model 3.

Source body	Corners of top surface(m) (Easting, Northing)	Top depth (m)	Bottom depth(m)
1	(900, 2100), (2000, 2100), (2000, 2500), (900, 2500)	200	700
2	(1000, 500), (1700, 500), (1700, 1600), (1400, 1600), (1400, 800)	200	500
3	(2700, 1500), (3200, 1500), (3200, 2100), (2700, 2100)	300	600
4	(2400, 700), (2800, 700), (2800, 1100), (2400, 1100)	200	700

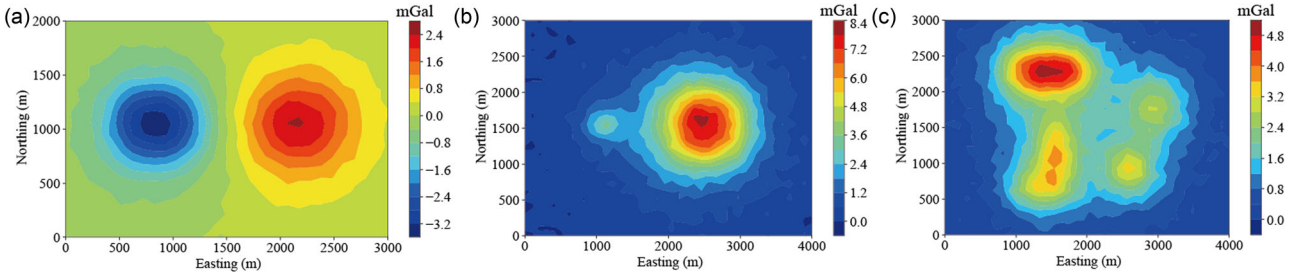


FIGURE 3 Forward gravity anomaly of synthetic Models 1–3 (a–c) with 2% Gaussian noise.

TABLE 2 Inversion parameters for synthetic tests.

Model no.	Range of density contrast(g/cm^3)	Momentum vector		Leapfrog parameter		Focusing parameter
		Mean	std	Step no.	Step size	
1	[−1,1]	0	0.001	0.005	[10, 50]	0.0001
2	[0, 1]	0	0.001	0.01	[10, 30]	0.01
3	[0, 1]	0	0.001	0.001	[10, 50]	0.001

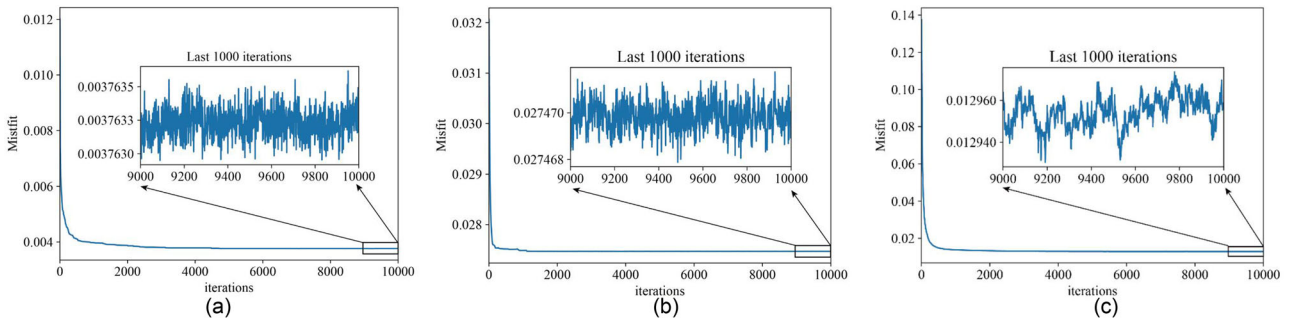


FIGURE 4 The misfit curve of Hamiltonian Monte Carlo (HMC) inversion of synthetic Models 1–3 (a–c).

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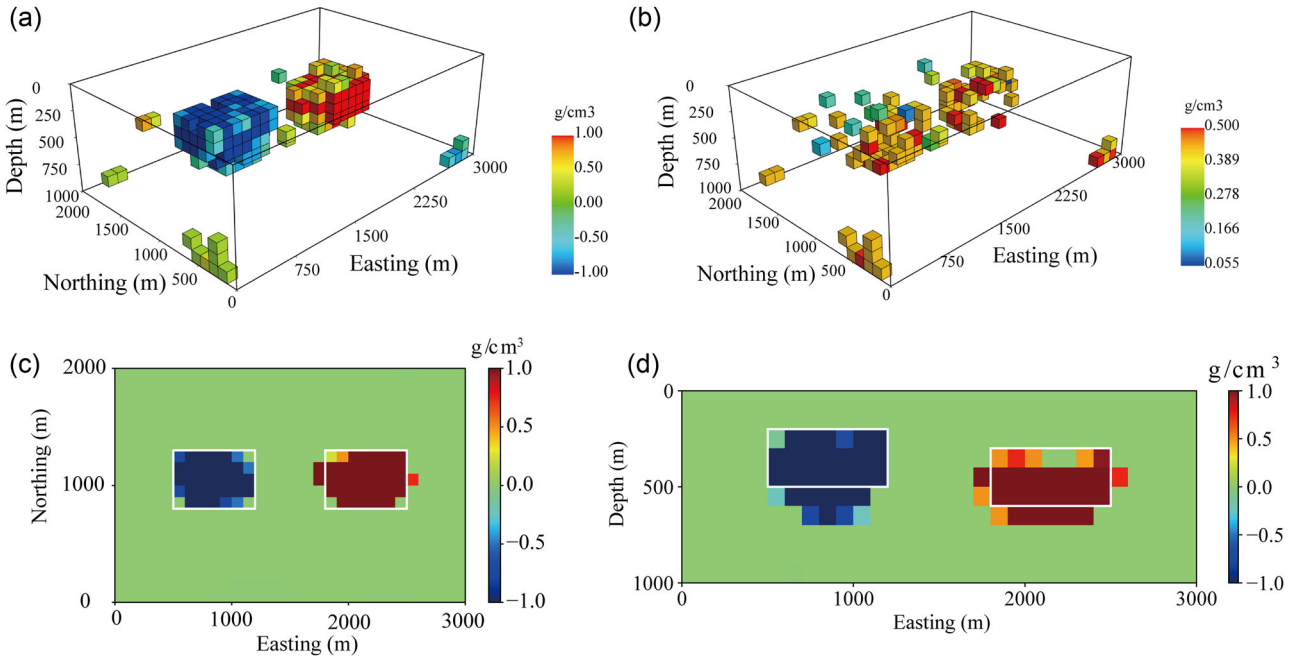


FIGURE 5 Inversion results of Model 1. (a) The average of the last 1000 iteration results as the final model. (b) The standard derivations of the last 1000 iterations. (c) Plane section at depth 400 m of inversion Model 1. (d) Cross-section at Northing = 1000 m of inversion Model 1. The white box in (c) and (d) indicates the true model position.

TABLE 3 Root-mean-square data (RMSD) and root-mean-square model (RMSM) of three synthetic model inversion tests.

Synthetic model no.	RMSD (mGal)	RMSD/gravity anomaly amplitude	RMSM (g/cm^3)
1	0.0454	1.89%	0.1035
2	0.1554	1.94%	0.1091
3	0.1012	1.94%	0.1167

TABLE 4 Comparison of the depth of the cap rock of the Vinton salt dome.

References	Top of Vinton	Bottom of Vinton (m)
Thompson and Eichelberger (1928)	200 m (south)–315 m(north)	600
Oliveira and Barbosa (2013)	/	460
Geng et al. (2014)	320 m	650
Qin and Huang (2016)	200 m (south)	600
This study	250 m	550

Here, we express the likelihood function and the prior model distribution with the data misfit and the model regularization term as probability functions (Menke, 1989):

$$\mathbf{P}(\mathbf{d}|\mathbf{m}) \propto \exp\left(\phi_d(\mathbf{m})\right), \tag{2}$$

$$\mathbf{P}(\mathbf{m}) \propto \exp\left(\varphi_s(\mathbf{m})\right). \tag{3}$$

The data misfit $\phi_d(\mathbf{m})$ is usually defined as (Vatankhah et al., 2023)

$$\phi_d(\mathbf{m}) = \|(\mathbf{Am} - \mathbf{d})\|_2^2. \tag{4}$$

The model regularization function $\phi_s(\mathbf{m})$ used is the minimum support function that provides the model with the minimum volume in the presence of an abnormal distribution of parameters, thus simplifying the model in a more compact area (Last & Kubik, 1983). The detailed regularization

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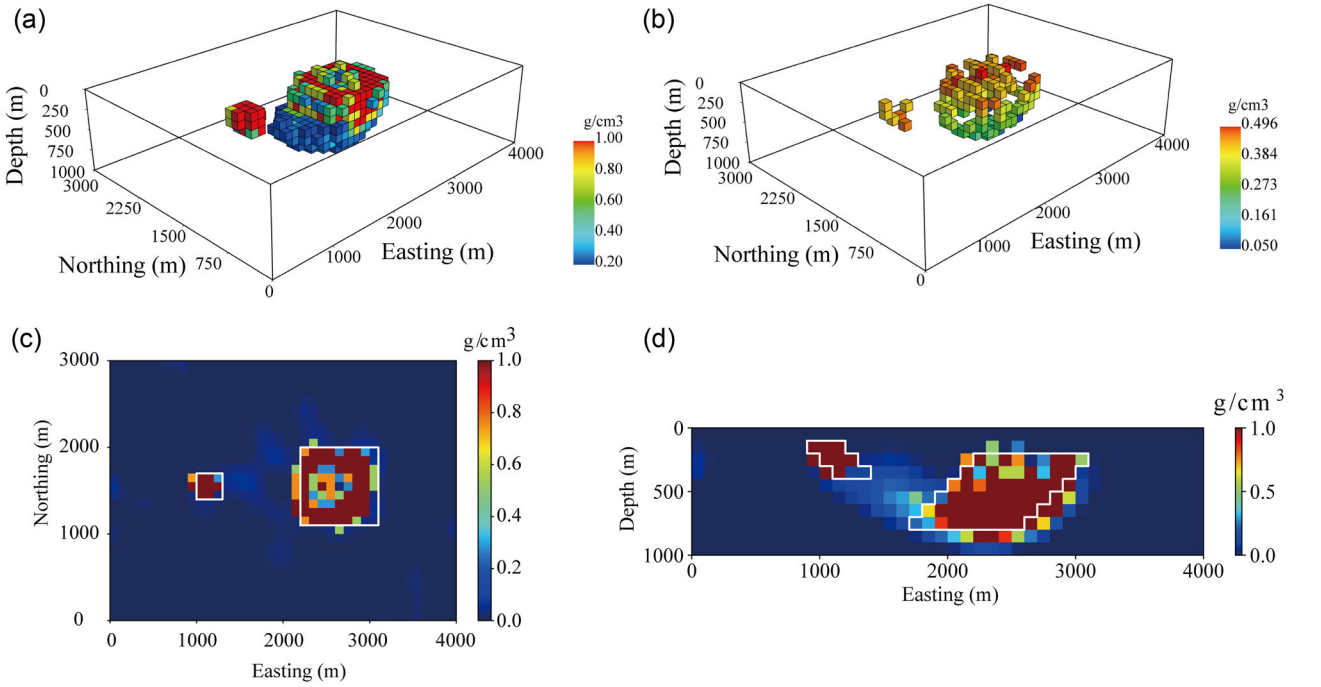


FIGURE 6 Inversion results of Model 2. (a) The average of the last 1000 iteration results as the final model. (b) The standard derivations of the last 1000 iterations. (c) Plane section at depth 200 m of inversion model 2. (d) Cross-section at Northing = 1500 m of inversion Model 2. The white polygon in (c) and (d) indicates the true model position.

function (Portniaguine & Zhdanov, 1999) here is as follows:

$$\varphi_s(\mathbf{m}) = \int_V \frac{(\mathbf{m}-\mathbf{m}^{\text{apr}})^2}{(\mathbf{m}-\mathbf{m}^{\text{apr}})^2+\beta}\mathrm{d}v, \tag{5}$$

where $\mathbf{m}^{\text{apr}}$ is a prior model and β is the focusing parameter ($0 < \beta \ll 1$) to ensure that the function is invertible.

Integrating equation with (1)–(5), we defined the objective function formulation by taking the negative natural logarithm of the posterior distribution:

$$J(\mathbf{m}) = -\log(\mathbf{P}(\mathbf{m}|\mathbf{d})) = (\mathbf{Am}-\mathbf{d})^{\mathrm{T}}\mathbf{C}_{\mathbf{D}}^{-1}(\mathbf{Am}-\mathbf{d}) + \alpha\sum_{i=1}^{N_m}\frac{(m_i - m_i^{\text{apr}})^2}{(m_i - m_i^{\text{apr}})^2+\beta}, \tag{6}$$

where $\mathbf{C}_{\mathbf{D}}^{-1}$ is the data covariance matrix, m_i is one element of the model parameters and $m_i{}^{\text{apr}}$ is one element of the prior model, α is a positive scalar as a parameter that balances the relative weighting between the two terms.

Hamiltonian Monte Carlo method for gravity inversion

The posterior distribution is typically addressed using Markov Chain Monte Carlo (MCMC) methods, which generate samples from the posterior distribution to construct a Markov chain. The stationary distribution of this chain aligned with the posterior distribution by accepting samples with a certain probability. However, all MCMC methods grapple with the curse of dimensionality in varying degrees (Gebraad et al., 2020). For instance, the classical Metropolis-Hasting algorithm (Metropolis et al., 1953; Mosegaard & Tarantola, 1995) sacrifices model exploration efficiency in exploring the model space to uphold high acceptance as dimensions increase, resulting in subsequent samples exhibiting high correlation. Hamiltonian Monte Carlo (HMC), leveraging derivatives of the target probability density with respect to the model parameters, can markedly progress from the current model, ensuring model independence while upholding high acceptance rates.

The HMC method is inspired by the Hamiltonian mechanic that describes the nature of moving particles in physics with energy conservation. The system can be described by a function of the position vector $\mathbf{q}$ and momentum vector $\mathbf{p}$ known as the Hamiltonian energy, $H(\mathbf{q},\mathbf{p})$ (Neal, 2011):

$$H(\mathbf{q},\mathbf{p}) = U(\mathbf{q}) + K(\mathbf{p}), \tag{7}$$

which consist of potential energy $U(\mathbf{q})$ and kinetic energy $K(\mathbf{p})$. In Hamiltonian dynamics, $\mathbf{q}$ can be seen as a position vector of particles that move in a high-dimensional space along some trajectory controlled by Hamilton's equation (Neal, 2011):

$$\frac{\partial q_i}{\partial t} = \frac{\partial H}{\partial p_i},\ \frac{\partial p_i}{\partial t} = -\frac{\partial H}{\partial q_i},\ i = 1,\ldots,n. \tag{8}$$

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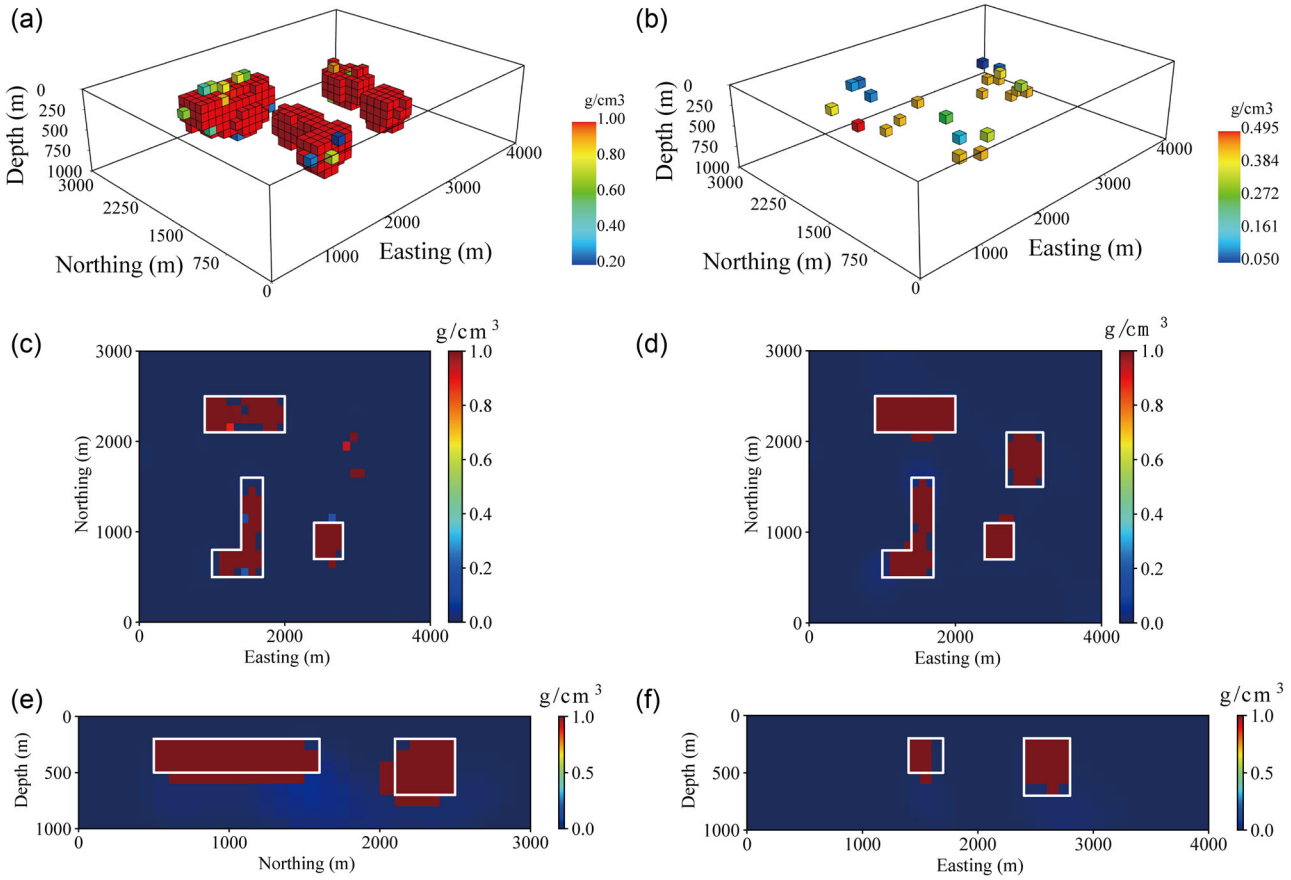


FIGURE 7 Inversion results of Model 3. (a) The average of the last 1000 iteration results as the final model. (b) The standard derivations of the last 1000 iterations. (c) Plane section at depth 200 m of inversion model 3. (d) Plane-section at depth 400 m of inversion Model 3. (e) Cross-section at Northing = 900 m of inversion Model 3. (f) Cross-section at Easting = 1500 m of inversion model 3. The white polygon in (c)–(f) indicates the true model position.

A notable characteristic of HMC dynamics is its preservation of the Hamiltonian, wherein the potential energy $U(\mathbf{q})$ and the kinetic energy $K(\mathbf{p})$ of the particles vary in opposite directions along the trajectory (Neal, 2011). With changes in the momentum vector $\mathbf{p}$ at different positions, particles traverse the entire model space, elucidating the fundamental principle behind HMC sampling.

Here, we integrate Hamiltonian energy into gravity inversion. In practice, $\mathbf{m}$ denotes the target model parameter, and $\mathbf{p}$ is an artificially introduced vector. The potential energy $U(\mathbf{m})$ can be represented by the negative natural logarithm of the posterior distribution as Equation (6), whereas the kinetic energy can be constructed based on Hamiltonian mechanics with a symmetric and positive definite mass matrix $\mathbf{M}$ as (Fichtner & Simute, 2018; Fichtner et al., 2019)

$$U(\mathbf{m}) = J(\mathbf{m}),\ K(\mathbf{p}) = \frac{\mathbf{p}^T\mathbf{M}^{-1}\mathbf{p}}{2}, \tag{9}$$

$$H(\mathbf{m},\mathbf{p}) = U(\mathbf{m}) + K(\mathbf{p}). \tag{10}$$

Sampling the new probability distribution $H(\mathbf{m},\mathbf{p})$ according to Hamilton's equation, we can get a new model $\mathbf{m}^*$ and a new momentum $\mathbf{p}^*$. This proposed state $(\mathbf{m}^*,\ \mathbf{p}^*)$ is accepted as the next state of the Markov chain with probability (Neal, 2011):

$$\prod_{\text{accept}} = \min\left[1, \exp\left(-H\left(\mathbf{m}^*,\ \mathbf{p}^*\right) + H\left(\mathbf{m},\mathbf{p}\right)\right)\right]. \tag{11}$$

Hence, the proposed state $(\mathbf{m}^*,\ \mathbf{p}^*)$ that increases the posterior will always be accepted; otherwise, it will be accepted with a random probability. Theoretically, the acceptance rate is $\Pi_{\text{accept}} = 100\%$ because the Hamiltonian is conserved (Neal, 2011). As $\mathbf{p}$ and $\mathbf{m}$ are independent random variables, we drop the momentum vector $\mathbf{p}$ and only keep the component we are interested in. That is, only the sampling of the posterior distribution is retained.

There is no analytical solution for Hamilton's equation, so the leapfrog method is proposed to solve Hamilton's equations

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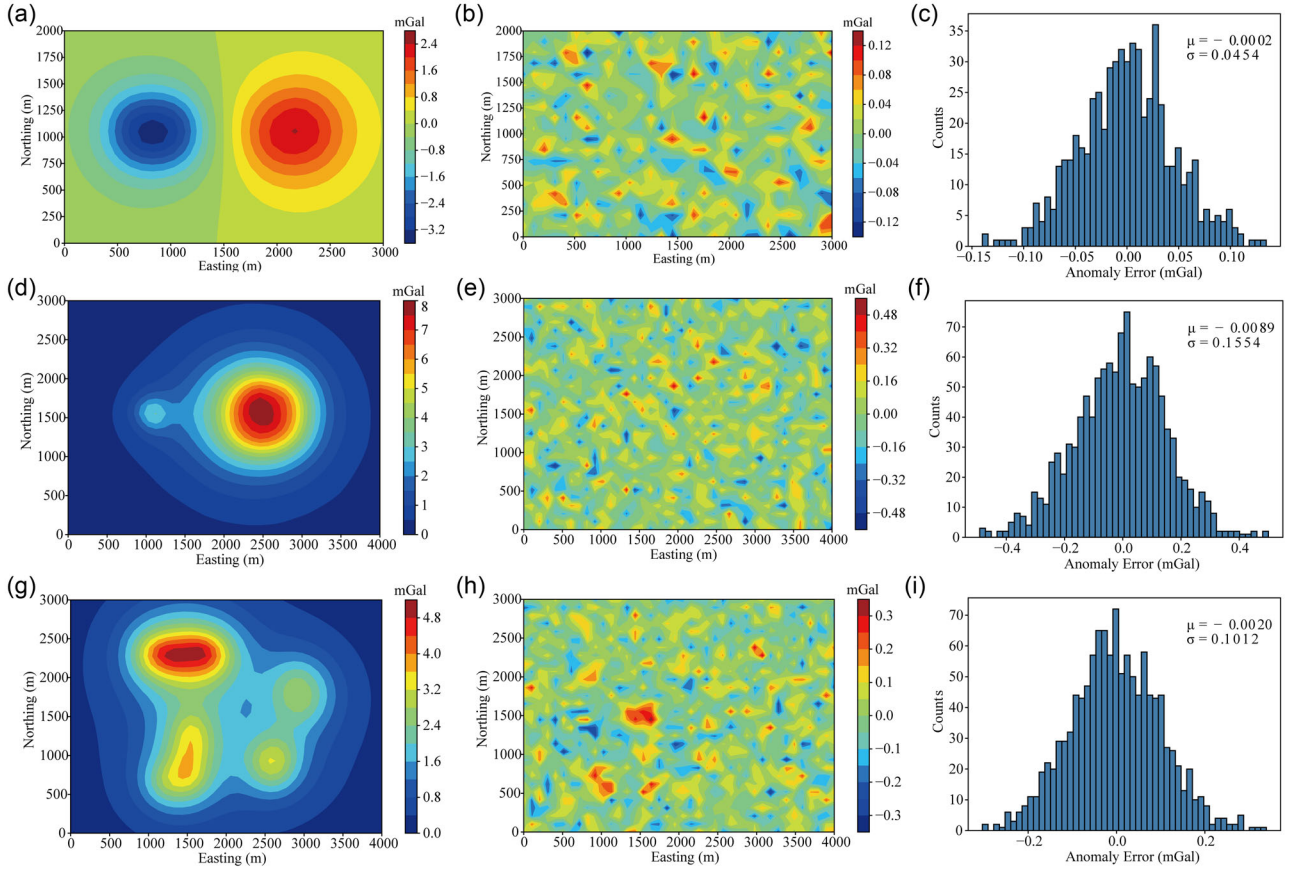


FIGURE 8 Predicated gravity anomaly, predicted error and its histogram for three synthetic models. (a–c) Model 1; (d–f) Model 2; (g–i): Model 3.

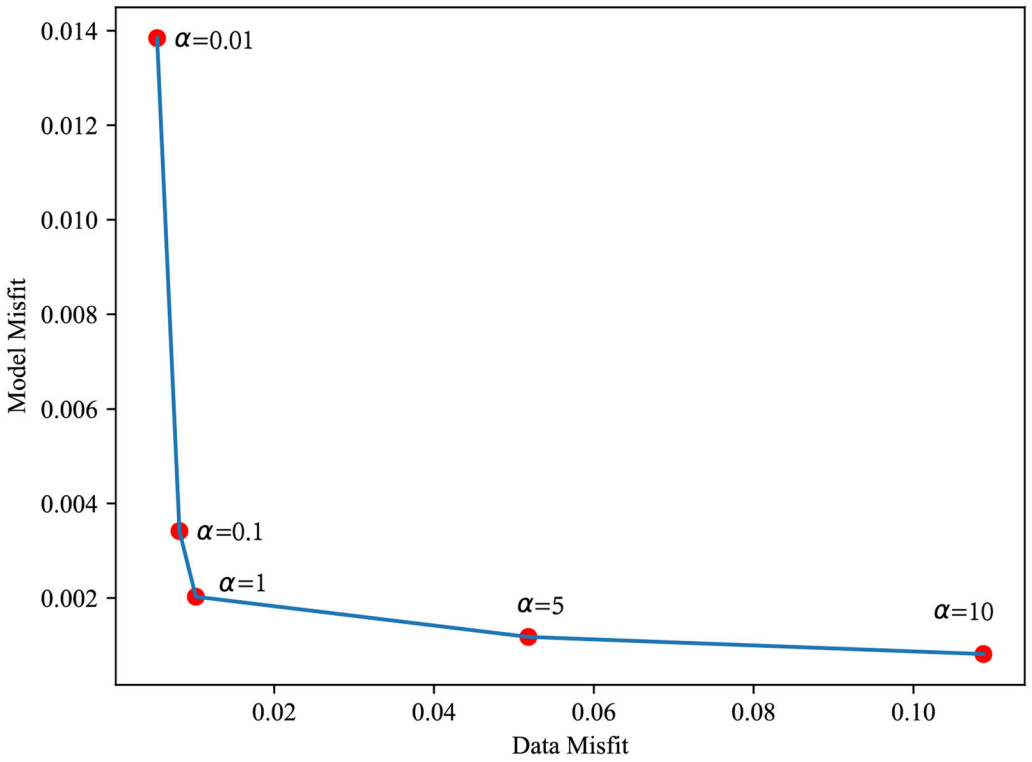


FIGURE 9 *L*-curve for regularization parameter test in Hamiltonian Monte Carlo (HMC) inversion of synthetic test 3.

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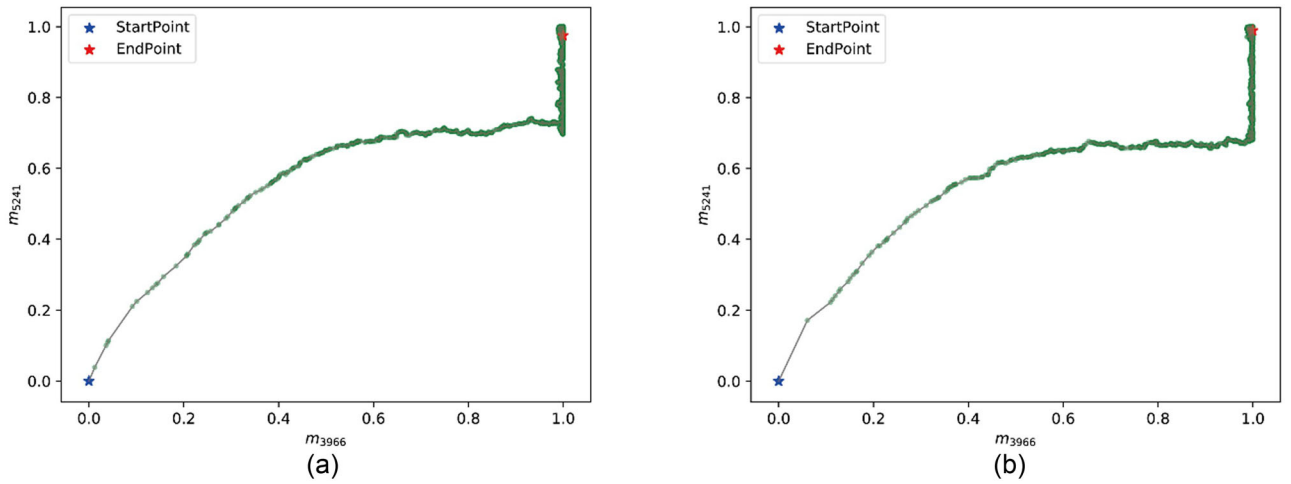


FIGURE 10 Hamiltonian Monte Carlo (HMC) trajectory of a pair of random model parameters with different mass matrix. Mass matrix defined by (a) unit matrix and (b) forward kernel matrices.

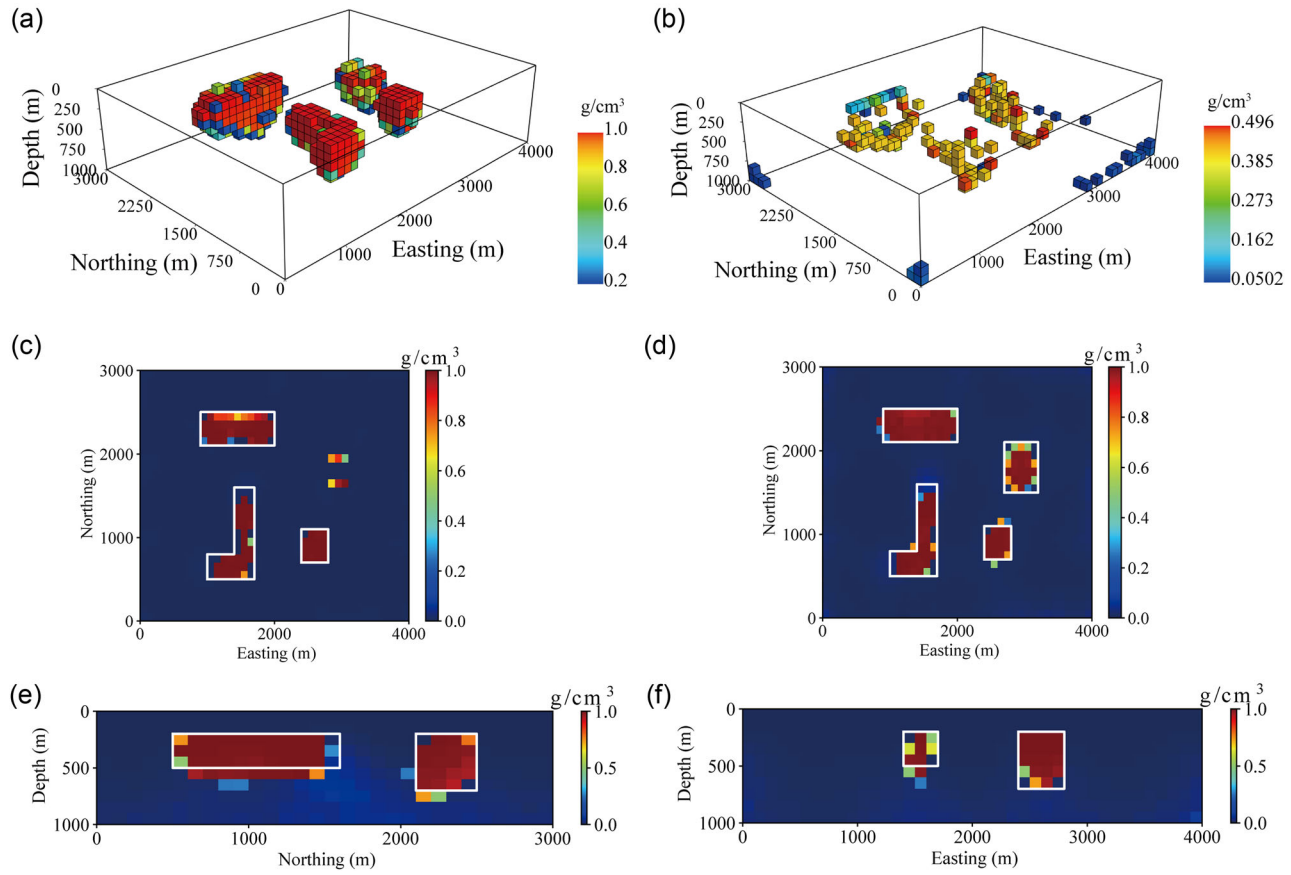


FIGURE 11 Inversion results of Model 3 using mass matrix of forward gravity kernel. (a) The average of the last 1000 iteration results as the final model. (b) The standard derivations of the last 1000 iterations. (c) Plane section at depth 200 m of inversion Model 3. (d) Plane-section at depth 400 m of inversion Model 3. (e) Cross-section at Northing = 900 m of inversion Model 3. (f) Cross-section at Easting = 1500 m of inversion Model 3. The white polygon in (c)–(f) indicates the true model position.

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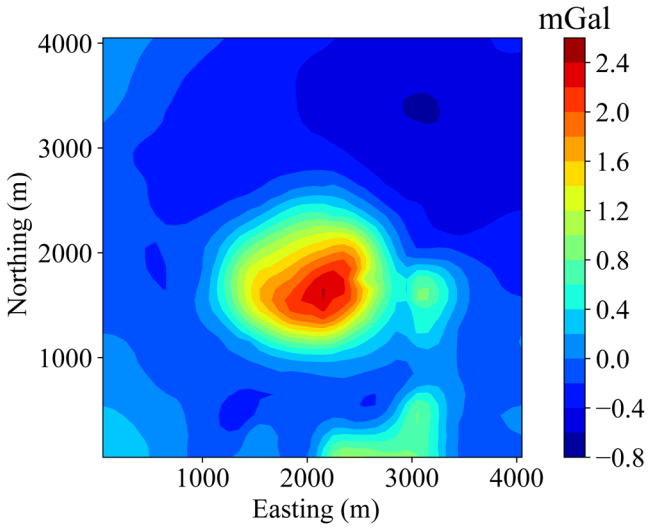


FIGURE 12 The gravity anomaly map of the Vinton salt dome (according to Gao et al., 2019).

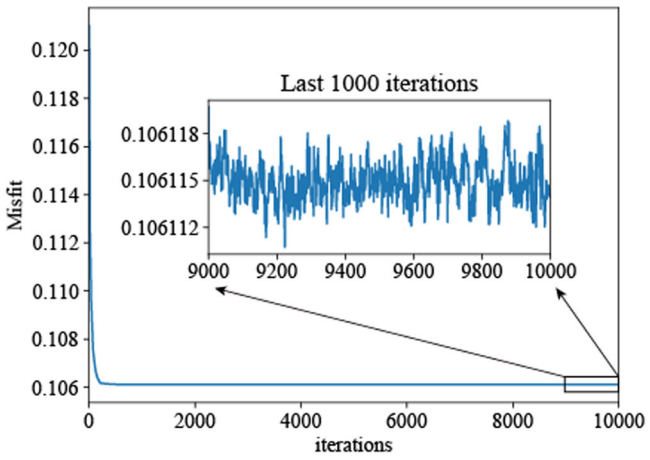


FIGURE 13 Misfit curve of Hamiltonian Monte Carlo (HMC) inversion applied to the Vinton salt dome.

numerically (Neal, 2011):

$$\mathbf{p}_i\left(t+\frac{\epsilon}{2}\right)=\mathbf{p}_i\ (t)-\left(\frac{\epsilon}{2}\right)\left.\frac{\partial U}{\partial m_i}\right|_t,$$

$$m_i\,(t+\epsilon)=m_i\ (t)+\epsilon\left.\frac{\partial U}{\partial m_i}\right|_{t+\epsilon/2}, \tag{12}$$

$$\mathbf{p}_i\,(t+\epsilon)=\mathbf{p}_i\ (t+\epsilon/2)-(\epsilon/2)\left.\frac{\partial U}{\partial m_i}\right|_{t+\epsilon},$$

where t is the discrete time, ϵ is the step size, and L is the number of steps. According to the leapfrog method, given the initial value $\mathbf{m}_i(0)$, $\mathbf{p}_i(0)$, ϵ and L, we can get the state at time ϵ, 2ϵ $3\epsilon,\ldots,$ $L\cdot\epsilon$. This is a trajectory of position and momentum, and the state with trajectory length $L\cdot\epsilon$ is the next sampling point. Solving Hamilton's equations numerically causes an accumulation of errors, which leads to $\Pi_{\text{accept}}\leq 100\%$. Notably, the numerical error approaches zero when the step size ϵ approaches zero. Furthermore, a small L leads to random walk sampling, whereas a large L may waste computational resources. Therefore, L and ϵ need to be appropriately tuned to obtain independent samples and an optimized acceptance rate with a reasonable amount of computation.

Algorithm implementation

In practice, due to the rapid decay of gravity anomaly with increasing depths, the depth weighting is usually used in gravity inversion to overcome the depth attenuation effects and provide higher vertical resolution. Here we used the model weighting function given by Zhdanov (2002)

$$\mathbf{W}_{\text{m}}=\text{diag}\left(\mathbf{A}^{\text{T}}\mathbf{A}\right)^{1/2}, \tag{13}$$

and the weighted model parameter $\tilde{\mathbf{m}}$ is

$$\tilde{\mathbf{m}}=\mathbf{W}_{\text{m}}\,\mathbf{m} \tag{14}$$

Integrated with the above HMC method, we set the potential energy $U(\tilde{\mathbf{m}})$ as Equation (6). The derivative of $U(\tilde{\mathbf{m}})$ with respect to $\tilde{\mathbf{m}}$ used in the Leapfrog method can be given in the following form:

$$\frac{\partial U}{\partial\tilde{\mathbf{m}}}=2\mathbf{A}^{\text{T}}\mathbf{C}_{\mathbf{D}}^{-1}\left(\mathbf{A}\tilde{\mathbf{m}}-\mathbf{d}\right)+\alpha\sum_{i=1}^{N_m}\frac{2\beta\left(\tilde{m}_i-\tilde{m}_i^{\text{apr}}\right)}{\left(\left(\tilde{m}_i-\tilde{m}_i^{\text{apr}}\right)^2+\beta\right)^2}. \tag{15}$$

A specific algorithm flow of the HMC gravity inversion is illustrated in Figure 1. The main process can be implemented by following steps:

(i) Given the current samples $\mathbf{m}_{\text{cur}}$ and transferred it to $\tilde{\mathbf{m}}_{\text{cur}}=\mathbf{W}_{\text{m}}\,\mathbf{m}_{\text{cur}}$, draw $\mathbf{p}_{\text{cur}}$ from the Gaussian distribution with mean $\mu=0$ and variance $\sigma=\mathbf{S}$; calculate the Hamiltonian energy with potential energy $U(\tilde{\mathbf{m}}_{\text{cur}})$ and kinetic energy $K(\mathbf{p}_{\text{cur}})$.

(ii) Obtain the proposed samples $(\tilde{\mathbf{m}}^*,\mathbf{p}^*)$ by simulating the Hamiltonian system using the Leapfrog method. In each step of the Leafprog process with a step size of ϵ, first do a half-step momentumm update: $\mathbf{p}_{\text{new}}^{\text{half}}=\mathbf{p}_{\text{cur}}-\epsilon/2\nabla(U(\tilde{\mathbf{m}}_{\text{cur}}))$, second the full-step position update: $\tilde{\mathbf{m}}_{\text{new}}=\tilde{\mathbf{m}}_{\text{cur}}+\epsilon\nabla K(\mathbf{p}_{\text{new}})$, then again a half-step momentum update:

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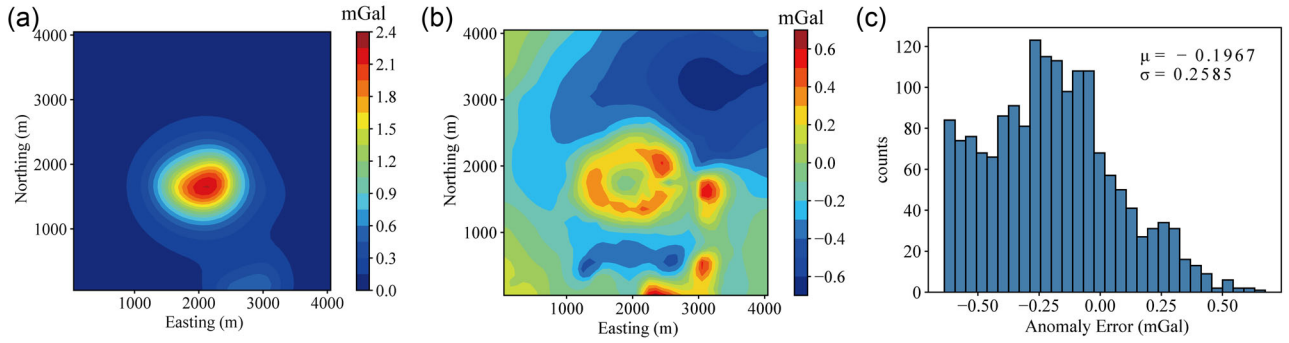


FIGURE 14 Predicted gravity anomaly of the Vinton salt dome (a), the difference between predicate gravity and observed gravity in Vinton salt dome (b) and its histogram map (c).

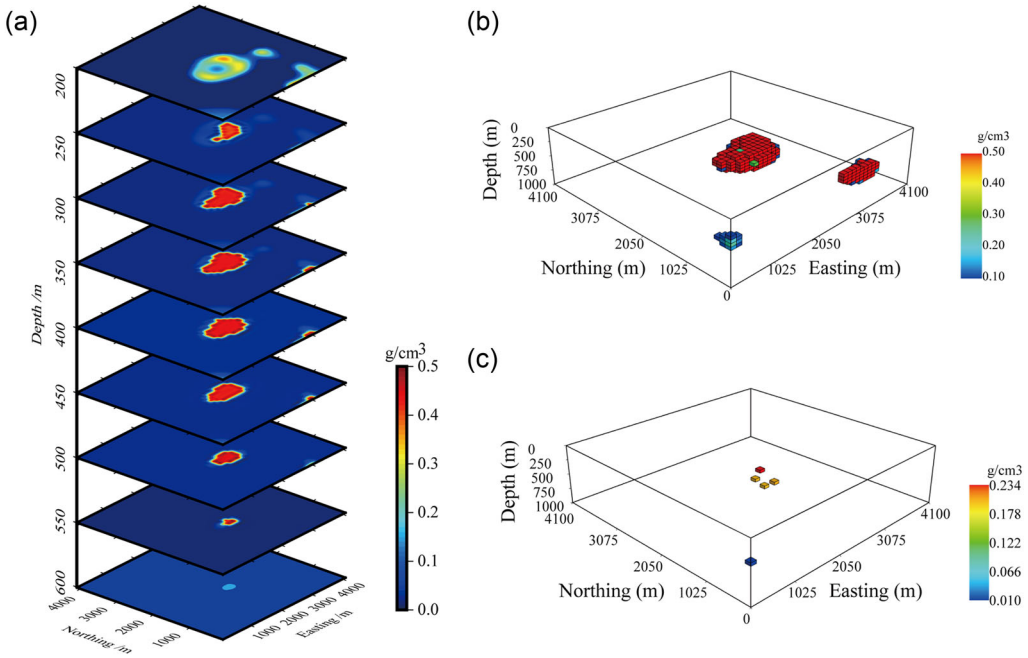


FIGURE 15 Density structure of Vinton salt dome from Hamiltonian Monte Carlo (HMC) gravity inversion. (a) Plane-sections from depth 200 to 600 m. (b) The average value of the last 1000 models. (c) The standard derivations of the last 1000 models.

$\mathbf{p}_{\text{new}} = \mathbf{p}_{\mathbf{new}}^{\mathbf{half}} - \epsilon/2\nabla(U(\tilde{\mathbf{m}}_{\text{new}}))$. With L step of above process, we get the the proposed sample $(\tilde{\mathbf{m}}^*, \mathbf{p}^*)$.

(iii) Decide to accept or reject the proposed samples according to Π_{accept}. If 'yes', the above proposed sample is accepted as a new sample; if 'not', the next state is the same as the current state.

With the iterative process of the above steps, we can get an accepted HMC sample set for PDF estimation.

The pseudo code of the HMC inversion algorithm can be illustrated as

1: **procedure** Gravity inversion using HMC
2: **Input**: $\boldsymbol{m}_0$, ϵ, L, samples
3: $\tilde{\boldsymbol{m}}_{cur} = \boldsymbol{W}_m \, \boldsymbol{m}_0$
4: **while** i < samples **do**
5: draw $\boldsymbol{p}_{cur}$ from N (0,$\boldsymbol{S}$)
6: $\boldsymbol{p}_{new} = \boldsymbol{p}_{cur}$
7: $\tilde{\boldsymbol{m}}_{new} = \tilde{\boldsymbol{m}}_{cur}$
8: $U(\tilde{\boldsymbol{m}}_{cur})$, $\nabla U(\tilde{\boldsymbol{m}}_{cur})$ = -ln(posterior($\tilde{\boldsymbol{m}}_{cur}$))
9: $K(\boldsymbol{p}_{cur}) = \frac{p_{cur}^T \boldsymbol{M}^{-1} p_{cur}}{2}$
10: $H(\tilde{\boldsymbol{m}}_{cur}, \boldsymbol{p}_{cur}) = U(\tilde{\boldsymbol{m}}_{cur})+K(\boldsymbol{p}_{cur})$
11: $\boldsymbol{p}_{new} = \boldsymbol{p}_{new} - \frac{\epsilon}{2}\nabla(U(\tilde{\boldsymbol{m}}_{cur}))$
12: **for** $l = 1 \rightarrow L$ **do**
13: $\tilde{\boldsymbol{m}}_{new} = \tilde{\boldsymbol{m}}_{new} + \epsilon\nabla K(\boldsymbol{p}_{new})$
14: $U(\tilde{\boldsymbol{m}}_{new})$, $\nabla U(\tilde{\boldsymbol{m}}_{new})$ = -ln(posterior($\tilde{\boldsymbol{m}}_{new}$))
15: **if** $l < L$ **then**
16: $\boldsymbol{p}_{new} = \boldsymbol{p}_{new} - \epsilon\nabla(U(\tilde{\boldsymbol{m}}_{new}))$
17: **else**
18: $\boldsymbol{p}_{new} = \boldsymbol{p}_{new} - \frac{\epsilon}{2}\nabla(U(\tilde{\boldsymbol{m}}_{new}))$

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19: **end if**
20: **end for**
21: $\boldsymbol{p}_{new} = -\boldsymbol{p}_{new}$
22: $K(\boldsymbol{p}_{new}) = \frac{\boldsymbol{p}_{new}^T \boldsymbol{M}^{-1} \boldsymbol{p}_{new}}{2}$
23: $H(\tilde{\boldsymbol{m}}_{new}, \boldsymbol{p}_{new}) = U(\tilde{\boldsymbol{m}}_{new}) + K(\boldsymbol{p}_{new})$
24: **if** randnum $\leq \exp(-(H(\tilde{\boldsymbol{m}}_{new}, \boldsymbol{p}_{new}) - H(\tilde{\boldsymbol{m}}_{cur}, \boldsymbol{p}_{cur}))$ **then**
25: $\tilde{\boldsymbol{m}}_{cur} = \tilde{\boldsymbol{m}}_{new}$
26: i = i + 1
27: **else**
28: $\widetilde{\boldsymbol{m}}_{cur} = \tilde{\boldsymbol{m}}_{cur}$
29: **end if**
30: **end while**
31: **Output**: $\boldsymbol{m} = \boldsymbol{W}_{\mathrm{m}}^{-1} \tilde{\boldsymbol{m}}_{cur}$
32: **end procedure**

SYNTHETIC TESTS

Synthetic models

To assess the effectiveness of the Hamiltonian Monte Carlo (HMC) algorithm, we devised three models characterized by distinct geometries and density contrasts (Figure 2). In all the synthetic tests, the observed gravity stations were uniformly distributed within a rectangular area, with intervals of 100 m in both x and y directions. Each model cell was discrete, spanning 100 m in each dimension.

The first model (Figure 2a) consists of two cubic bodies with the same dimensions of 500 × 700 × 300 m^3. These bodies are positioned at depths of 200 and 300 m, with density contrasts of −1 and 1 g/cm^3 relative to the background, respectively.

In the second model (Figure 2b), two dipping dykes of differing sizes and burial depths are depicted. The smaller dyke on the left side extends from 100 to 400 m, whereas the larger one on the right spans from 200 to 800 m vertically. The inclinations of these dykes are 135 and 45°, respectively, with identical density contrasts of 1 g/cm^3.

The third mode, illustrated in Figure 2c, is a complex configuration featuring four bodies of varying geometries, sizes, and depths. The geometric parameters of these bodies are detailed in Table 1, with corner points arranged counterclockwise from the bottom left corner. Each of these bodies has a density contrast of 1 g/cm^3.

The gravity anomaly of three synthetic models is calculated and a Gaussian noise with a mean of zero and a variance of 2% of the gravity anomaly amplitude of coressponding model is added to the synthetic gravity data (Figure 3).

Test and results

The HMC inversion algorithm was applied to the three synthetic datasets and the inversion results were analysed. An initial model with zero density was provided for any of the tests. The inversion parameters, including the momentum vector **p** following a Gaussian distribution with a mean (μ) and standard derivation (σ), as well as the number of steps (L) and step size (ϵ) utilized in the Leapfrog stage and the focusing parameter (β) for the model function, were set based on empirical experiments. The inversion parameters for tests are listed in Table 2.

We conducted 10,000 samplings in HMC inversion for each synthetic model. The misfits of three model inversions are depicted in Figure 4, showing rapid convergence to a stable level in all synthetic tests. The warm-up phase for HMC mostly lasted less than 1000 iterations. The acceptance ratio reached 100% for Models 1 and 2 and 99.99% for Model 3. Subsequently, the stability of the inversion is indicated by the statistics of the last 1000 iterations.

The mean and standard deviation of the last 1000 successive simulations were calculated to determine the optimal inversion result and its uncertainty. Plane- and cross-sections of the inversion results aided further analysis. Figures 5–7 illustrate the inversion results of synthetic Models 1–3, respectively. Figure 8 displays the predicted gravity anomaly, the predicted gravity error and its statistical histogram map for each synthetic test.

From the inversion results (Figures 5–8), it is evident that all synthetic tests yielded favourable outcomes with very small uncertainty in each inversion model. The inversion models effectively reconstructed the shape of the synthetic models, with the predicted gravity error also being minimal for each synthetic test. The statistical histograms indicate that the predicted gravity error predominantly clusters near zero with a small standard deviation.

Further evaluation of the inversion results was performed using root-mean-square data misfit (RMSD) and root-mean-square model recovery (RMSM). These two indicators are defined as

$$\begin{aligned} \mathrm{RMSD} &= \sqrt{\frac{\sum_{i=1}^{N_d} \left(d_i^{\mathrm{for}} - d_i^{\mathrm{obs}}\right)^2}{N_d}} \\ \mathrm{RMSM} &= \sqrt{\frac{\sum_{j=1}^{N_m} \left(m_j^{\mathrm{inv}} - m_j^{\mathrm{mod}}\right)^2}{N_m}}. \end{aligned} \tag{13}$$

The analysis of the three model tests is summarized in Table 3 based on the inversion results from the last 1000 iterations. The RMSD and RMSM values for the three models are low, with the data fitting error being limited to 2% of the original gravity anomaly amplitude.

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DISCUSSION

The regularization parameter (α) plays a pivotal role in gravity inversion as it effectively balances the likelihood probability density function (PDF) and the prior PDF. The *L*-shaped curve with $\varphi_s(m)$ plotted as a function of $\phi_d(m)$ for all possible values of the regularization parameters is generally used for parameter selection. In this study, we explored various regularization parameters across a wide range and identified the optimal one for inversion. For instance, in Model 3, we tested the regularization parameter with values of 0.01, 0.1, 1, 10, 5 and 10. The variations of the data misfit and model misfit (*L*-curve) are depicted in Figure 9, revealing that setting $\alpha = 1$ yielded an optimal point for the inversion. As one of the key parameters, the regularization parameter can be estimated automatically by a cooling process based *L*-curve analysis (Vatankhah et al., 2014).

Regularization inversion also incorporated different stabilization terms for constraints, such as $L1$-norm or Lp-norm terms (Vatankhah et al., 2017; Wei & Sun, 2021; Vatankhah et al., 2023; Wei & Sun, 2023). The proposed Hamiltonian Monte Carlo (HMC) inversion framework can likewise accommodate various stabilization terms for the inversion. Here, the Minimum Support Stabilization employed has demonstrated an effective recovery of the models' boundaries. Tests with $L1$ norm and $L2$ norm stabilization reveal differences in the identification of the model edges, with $L2$ norm stabilization producing smoother results. Exploring different stabilization methods in HMC inversion can effectively recover the basic shape and location of the models.

The selection of the focusing parameter β is a key issue for the inversion. It has obvious influence on the smoothness of the model. Proper focusing parameter can get a clear description of sharp boundaries, but it may be unstable at different values of β (Zhdanov, 2009). Different values were tested to reach a reasonable model with sharper boundaries in our test.

Another critical parameter for HMC inversion is the selection of the mass matrix, which significantly influences sampling efficiency. Although the mass matrix can theoretically be any symmetric positive-definite matrix, practical consideration impacts its selection. Discussions on mass matrix definition range from using a unit matrix (as utilized in this paper) to employing geophysical forward kernel matrices (Fichtner & Simute, 2018) or other matrices related to data covariance and prior model covariance matrix (Fichtner et al., 2019). These kinds of definitions on the mass matrix are convenient for calculations where the forward equation is linear, and the prior is a Gaussian distribution. In this paper, the unit matrix shows good efficiency for gravity inversion. As a comparison, we tested the mass matrix defined with the gravity forward kernel in synthetic Model 3. Results indicate that sampling efficiently reaches the optimal target with models closely resembling (Figure 10), the true model (Figure 11), albeit with slightly larger uncertainties compared to using the unit matrix. This comparison underscores the applicability of simple definite mass matrices, whether unit or forward kernel-based, in linear gravity inversion.

The three synthetic models were tested on a Dell laptop equipped with a 2.3-GHz 11th Gen Intel(R) Core (TM) i7-11800H CPU and 16 GB of RAM. For 10,000 iterations, the computation time for the entire inversion ranges from 1236 to 7198 s, respectively. Our primary focus was on assessing the feasibility of the HMC inversion. However, there is potential for further optimization of the code to reduce iteration numbers and enhance efficiency. Notably, the inversion quickly converges, suggesting that iteration numbers can be further reduced to improve efficiency. Statistical analysis of inversion results for Model 3 at different intervals of iterations reveals minimal variation in data misfit and model misfit.

APPLICATION CASE STUDY

To establish the efficacy of the Hamiltonian Monte Carlo (HMC) algorithm in inversion, we applied this method to scrutinize the Vinton salt dome situated in Calcasieu Parish, Louisiana, aiming to unravel its subsurface geometry. This dome has been studied for over a century, dating back to the inaugural drill in May 1901 (Thompson & Eichelberger, 1928). As a quintessential Gulf Coast salt dome, extensive geophysical investigations have probed its structural framework (Constance et al., 1999). Comprising a central core of massive salt and a distinct cap rock rises above it (Coker et al., 2007). The dome's cap rock transitions from limestone to gypsum and anhydrite, boasting an average density of 2.75 g/cm^3. Surrounding sediments and the salt dome itself exhibit an average density of 2.2 g/cm^3 (Oliveira & Barbosa, 2013). Consequently, gravity-related methodologies have been extensively used to characterize the cap rock, revealing a substantial density contrast with the surrounding formations (Ennen & S. Hall, 2011; Oliveira & Barbosa, 2013; Geng et al., 2014; Qin & Huang, 2016; Gao et al., 2019).

Utilizing on the gravity anomaly after background removal (Gao et al., 2019; Figure 12), we applied HMC gravity inversion to dissect the three-dimensional (3D) distribution of the Vinton salt dome's cap rock. The study area was discretized into $41 \times 41 \times 20 = 33,620$ prisms, each measuring 100 m × 100 m × 50 m for inversion purposes. The density contrast ranged from 0 to 0.55 g/cm^3, derived from prior density information. For the HMC inversion in the Vinton salt dome, the set momentum vector $\mathbf{p}$ was governed by a Gaussian distribution with mean $\mu = 0$, $\sigma = 0.001$. Two parameters in Leapfrog are set as step size $\epsilon = 0.001$ and number of steps $L = [10, 50]$. The focusing parameter on the model function is set as $\beta = 0.001$. A total of 10,000 HMC

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samples were collected, and the misfit function is depicted in Figure 13. Notably, the HMC inversion swiftly attains detailed balance ensuring stability. The final fitting result, showcased in Figure 14, adeptly captures the gravity anomaly, with mean and standard deviation values of the fitting result in the last 1000 successive simulations standing at −0.1967 and 0.2585 mGal, respectively. The average density structure and standard deviation of the last 1000 successive simulations serve as the inversion result and its uncertainty, respectively, illustrated in Figure 15b,c. A 3D visualization reveals the meticulous recovery of the Vinton dome with minimal uncertainty. The slice map from 200 to 600 m effectively delineates the cap rock's shape, which extends from about 250 to 550 m, aligning with previous findings (Table 4).

CONCLUSION

The Hamiltonian Monte Carlo theory provides an efficient way for multi-dimensional sampling. In this paper, it was applied to gravity inversion to determine the subsurface density distribution in this paper. The posterior probability density function of proposed model as the potential energy and a constructed kinetic energy term formed a Hamiltonian dynamics system for the Hamiltonian Monte Carlo (HMC) inversion. The HMC inversion algorithm used the Leapfrog method to approximate the proposed Hamiltonian dynamics system. Three synthetic model tests show that HMC gravity inversion can recover the models efficiently and accurately. HMC inversion gives the optimal model with a mean value and the uncertainty of the inversion based on statistics of accepted results. The application inversion of gravity data of the Vinton salt dome illustrates that the cap rock mainly extends from 250 to 550 m. It is consistent with results from previous gravity and gravity tensor data inversions. The synthetic and real application experiments showed the HMC method can be an effective and practical inversion method for gravity inversion.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Ya Xu https://orcid.org/0000-0002-7649-1737

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